

Problem 11.24

An investor deposits 1,000 on January 1 of year x and deposits 1,000 on January 1 of year $x + 2$ into a fund that matures on January 1 of year $x + 4$. The interest rate on the fund differs every year and is equal to the annual effective rate of growth of the gross domestic product (GDP) during the fourth quarter of the previous year.

The following are the relevant GDP values for the past four years:

Year	$x - 1$	x	$x + 1$	$x + 2$
Quarter III	800.0	850.0	900.0	930.0
Quarter IV	808.0	858.5	918.0	948.6

What is the effective interest rate earned by the investor over the four-year period?

Solution

The fourth-quarter growth factor for each year is the ratio of the GDP in Quarter IV to the GDP in Quarter III. Since this is a quarterly growth factor, its corresponding annual effective accumulation factor is its fourth power. Thus,

$$\begin{aligned}F_1 &= \left(\frac{808.0}{800.0}\right)^4 = (1.01)^4, \\F_2 &= \left(\frac{858.5}{850.0}\right)^4 = (1.01)^4, \\F_3 &= \left(\frac{918.0}{900.0}\right)^4 = (1.02)^4, \\F_4 &= \left(\frac{948.6}{930.0}\right)^4 = (1.02)^4.\end{aligned}$$

The first deposit accumulates for all four years, so its value at maturity is

$$\begin{aligned}1000F_1F_2F_3F_4 &= 1000(1.01)^8(1.02)^8 \\&\approx 1268.739217.\end{aligned}$$

The second deposit accumulates during the final two years, giving

$$\begin{aligned}1000F_3F_4 &= 1000(1.02)^8 \\&\approx 1171.659381.\end{aligned}$$

Therefore, the total value at maturity is

$$1268.739217 + 1171.659381 = 2440.398598.$$

Let j be the annual effective rate earned by the investor. The first deposit earns this rate for four years and the second for two years. Hence,

$$1000(1+j)^4 + 1000(1+j)^2 = 2440.398598.$$

Let $y = (1+j)^2$. Then

$$\begin{aligned} y^2 + y &= 2.440398598, \\ y &= \frac{-1 + \sqrt{1 + 4(2.440398598)}}{2} \\ &\approx 1.140243457. \end{aligned}$$

Consequently,

$$\begin{aligned} j &= \sqrt{y} - 1 \\ &\approx 0.0678218283. \end{aligned}$$

Thus, the annual effective rate earned over the four-year period is

$$\boxed{j \approx 6.7822\%}.$$